How Nafi Members Study!

1. Please subscribe to our YouTube channel and click the bell icon to get alerts https://www.youtube.com/channel/UC2yAW4Oq27r1yuRE8ePKRvA
2. Follow us on Facebook https://www.facebook.com/info.alnafi/
3. Follow us on Twitter https://twitter.com/nafiPakistan
4. Join the Al Nafi Official Group https://www.facebook.com/groups/alnafi/ (This group is only for members to ask questions)
5. All Nafi members MUST study on the portal https://alnafi.com/login/ and connect using your Nafi member username and password.
6. Once on the portal they can follow their classes by:
    - watching videos
    - attempting quizzes and flash cards
    - studying official Nafi notes
    - keep track of their studies along with many more features

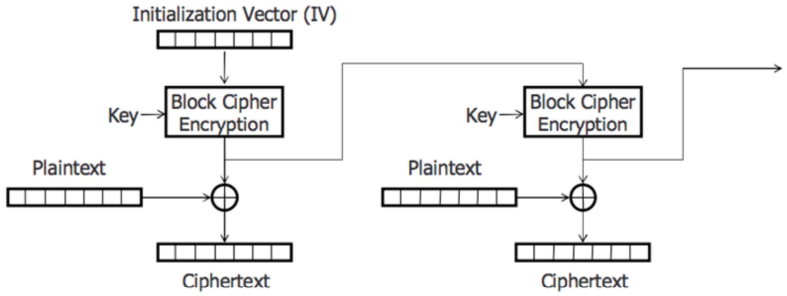
© 2018 Al-Nafi. All Rights Reserved.

Operations of Ciphers

- **Statistically unpredictable and unbiased:** Given that you know successive bits from the keystream, it would not possible to predict any of the following bits.
- **Key stream should be linearly related to the crypto varible:** which means that knowledge of the key stream output value does not disclose the crypto variable (encryption/decryption key).
- **Statistically unbiased:** There should be as many 0s as 1s and as many 00s as 01s, 10s, 11s, etc.
- **Functional complexity:** Each key stream bit should depend on most or all of the crypto variable bits.
- **Long periods:** without repetition.

© 2018 Al-Nafi. All Rights Reserved.

Block Ciphers



© 2018 Al-Nafi. All Rights Reserved. 3

What is a block cipher?

A block cipher is an encryption algorithm that encrypts a fixed size of n-bits of data - known as a block - at one time. The usual sizes of each block are 64 bits, 128 bits, and 256 bits. So for example, a 64-bit block cipher will take in 64 bits of plaintext and encrypt it into 64 bits of ciphertext. In cases where bits of plaintext is shorter than the block size, padding schemes are called into play. Majority of the symmetric ciphers used today are actually block ciphers. DES, Triple DES, AES, IDEA, and Blowfish are some of the commonly used encryption algorithms that fall under this group.

Popular block ciphers

DES - DES, which stands for Data Encryption Standard, used to be the most popular block cipher in the world and was used in several industries. It's still popular today, but only because it's usually included in historical discussions of encryption algorithms. The DES algorithm became a standard in the US in 1977. However, it's already been proven to be vulnerable to brute force attacks and other cryptanalytic methods. DES is a 64-bit cipher that works with a 64-bit key. Actually, 8 of the 64 bits in the key are parity bits, so the key size is technically 56 bits long.

3DES - As its name implies, 3DES is a cipher based on DES. It's practically DES that's run three times. Each DES operation can use a different key, with each key being 56 bits long. Like DES, 3DES has a block size of 64 bits. Although 3DES is many times stronger than DES, it is also much slower (about 3x slower). Because many organizations found 3DES to be too slow for many applications, it never became the ultimate successor of DES. That distinction is reserved for the next cipher in our list - AES.

AES - A US Federal Government standard since 2002, AES or Advanced Encryption Standard is arguably the most widely used block cipher in the world. It has a block size of 128 bits and supports three possible key sizes - 128, 192, and 256 bits. The longer the key size, the stronger the encryption. However, longer keys also result in longer processes of encryption. For a discussion on encryption key lengths, read Choosing Key Lengths for Encrypted File Transfers.

Blowfish - This is another popular block cipher (although not as widely used as AES). It has a block size of 64 bits and supports a variable-length key that can range from 32 to 448 bits. One thing that makes blowfish so appealing is that Blowfish is unpatented and royalty-free.

Twofish - Yes, this cipher is related to Blowfish but it's not as popular (yet). It's a 128-bit block cipher that supports key sizes up to 256 bits long.

Statistical Analysis

- Knowing % of occurrences of different letters (e.g. e occurs 13% of time in the documer 19% of times)
- Knowing commonly occurring two and three letter combinations (e.g. in, it, the, ion, ing
- If some knowledge about the content is available it is even easier to crack

Initialization Vectors (IVs) – Why They Are Needed

- Because the IV is a random starting point, or a random number, that starts the process, it ensures that we add complexity and randomness into the encryption process.

© 2018 Al-Nafi. All Rights Reserved.

Reference https://en.wikipedia.org/wiki/Initialization_vector

In cryptography, an **initialization vector** (**IV**) or **starting variable** (**SV**)[1] is a fixed-size input to a cryptographic primitive that is typically required to be random or pseudorandom. Randomization is crucial for encryption schemes to achieve semantic security, a property whereby repeated usage of the scheme under the same key does not allow an attacker to infer relationships between segments of the encrypted message. For block ciphers, the use of an IV is described by the modes of operation. Randomization is also required for other primitives, such as universal hash functions and message authentication codes based thereon.

Some cryptographic primitives require the IV only to be non-repeating, and the required randomness is derived internally. In this case, the IV is commonly called a nonce (*number used once*), and the primitives are described as *stateful* as opposed to *randomized*. This is because the IV need not be explicitly forwarded to a recipient but may be derived from a common state updated at both sender and receiver side. (In practice, a short nonce is still transmitted along with the message to consider message loss.) An example of stateful encryption schemes is

the counter mode of operation, which uses a sequence number as a nonce.

The size of the IV is dependent on the cryptographic primitive used; for block ciphers, it is generally the cipher's block size. Ideally, for encryption schemes, the unpredictable part of the IV has the same size as the key to compensate time/memory/data tradeoff attacks.[2][3][4][5] When the IV is chosen at random, the probability of collisions due to the birthday problem must be taken into account. Traditional stream ciphers such as RC4 do not support an explicit IV as input, and a custom solution for incorporating an IV into the cipher's key or internal state is needed. Some designs realized in practice are known to be insecure; the WEP protocol is a notable example, and is prone to related-IV attacks.

Kerckhoffs's Principle

Named after Dutch cryptographer Auguste Kerckhoffs, the principle states that **"A cryptosystem should be secure even if everything about the system, except the key, is public knowledge."** In other words, most people summarize that as meaning **"the enemy knows the system"** and, therefore, the only thing that provides security in cryptography is security of the key. **We have to assume that the enemy will know the methods and the algorithms**, so protecting the key become the most important aspect of cryptography.

© 2018 Al-Nafi. All Rights Reserved.

Reference https://en.wikipedia.org/wiki/Kerckhoffs%27s_principle

Read this article to build an understanding and only try to understand the concept and its definitions. That is the total scope you need to know for the exam.

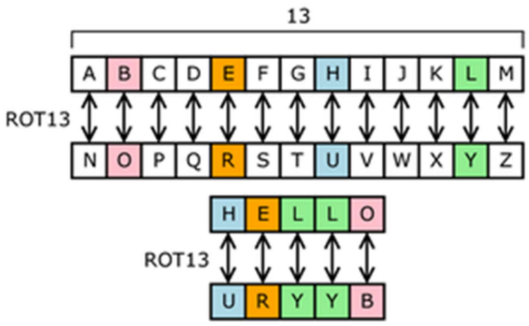
High Work Factor

- The average amount of effort or work required to break an encryption system is referred to as the work factor.

© 2018 Al-Nafi. All Rights Reserved.

The average amount of effort or work required to break an encryption system is referred to as the work factor. That is to say, decrypting a message without having the entire encryption key or to find a secret key given all or part of a ciphertext would also be referred to as the work factor of the cryptographic system. Typically, the work factor is measured in some units such as hours of computing time on one or more given computer systems or a cost in dollars of breaking the encryption. If the work factor is
sufficiently high, the encryption system is considered to be

Substitution Ciphers



© 2018 Al-Nafi. All Rights Reserved.

Reference https://en.wikipedia.org/wiki/Substitution_cipher

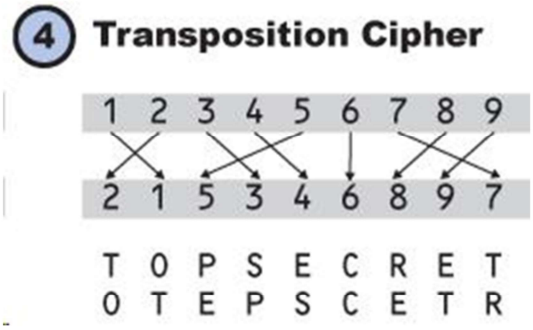
In cryptography, a **substitution cipher** is a method of encrypting by which units of plaintext are replaced with ciphertext, according to a fixed system; the "units" may be single letters (the most common), pairs of letters, triplets of letters, mixtures of the above, and so forth. The receiver deciphers the text by performing the inverse substitution.

Substitution ciphers can be compared with transposition ciphers. In a transposition cipher, the units of the plaintext are rearranged in a different and usually quite complex order, but the units themselves are left unchanged. By contrast, in a substitution cipher, the units of the plaintext are retained in the same sequence in the ciphertext, but the units themselves are altered.
There are a number of different types of substitution cipher. If the cipher operates on single letters, it is termed a **simple substitution cipher**; a cipher that operates on larger groups of letters is termed **polygraphic**. A **monoalphabetic cipher** uses fixed substitution over the entire message, whereas a **polyalphabetic cipher** uses a number of substitutions at different positions in the message, where a unit from

the plaintext is mapped to one of several possibilities in the ciphertext and vice versa.

Read the remaining article to build an understanding of substitution cipher.

Transposition Ciphers



© 2018 Al-Nafi. All Rights Reserved.

Reference https://en.wikipedia.org/wiki/Transposition_cipher

Read the article to build a good understanding of transposition cipher

In cryptography, a **transposition cipher** is a method of encryption by which the positions held by units of plaintext (which are commonly characters or groups of characters) are shifted according to a regular system, so that the ciphertext constitutes a permutation of the plaintext. That is, the order of the units is changed (the plaintext is reordered). Mathematically a bijective function is used on the characters' positions to encrypt and an inverse function to decrypt.

Monoalphabetic and Polyalphabetic Ciphers

Mono alphabetic substitution cipher

Consider we have the plain text "cryptography". By using the substitution table shown below, we can encrypt our plain text as follows

Plain	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Cipher	J	I	B	R	K	T	C	N	O	F	Q	Y	G	A	U	Z	H	S	V	W	M	X	L	D	E	P

one permutation of the possible 26!

plain text : c r y p t o g r a p h y

cipher text : B S E Z W U C S J Z N E

Hence we obtain the cipher text as "BSEZWUCSJZNE"

© 2018 Al-Nafi. All Rights Reserved.

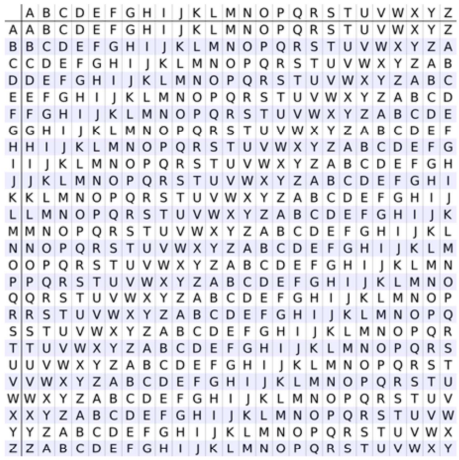
Reference https://vivadifferences.com/monoalphabetic-cipher-vs-polyalphabetic-cipher-5-basic-difference-plus-example/

Read this paper carefully as it will build your understanding of the topic in good detail.

Monoalphabetic Cipher. A monoalphabetic cipher is a substitution cipher in which the cipher alphabet is fixed through the encryption process. All of the substitution ciphers we have seen prior to this handout are monoalphabetic; these ciphers are highly susceptible to frequency analysis.

Polyalphabetic Cipher. A polyalphabetic cipher is a substitution cipher in which the cipher alphabet changes during the encryption process.

Polyalphabetic Ciphers



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

© 2018 Al-Nafi. All Rights Reserved.

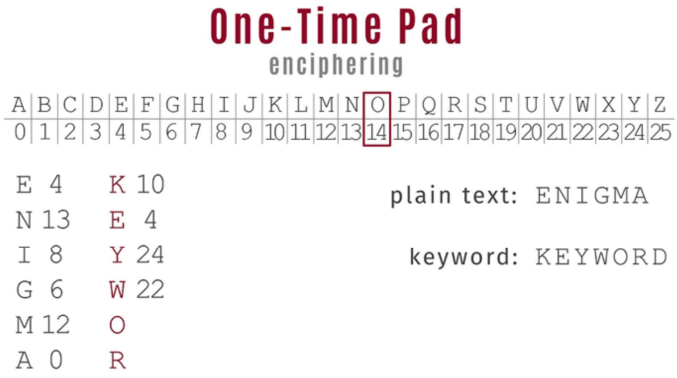
Reference https://vivadifferences.com/monoalphabetic-cipher-vs-polyalphabetic-cipher-5-basic-difference-plus-example/

Read this paper carefully as it will build your understanding of the topic in good detail.

Monoalphabetic Cipher. A monoalphabetic cipher is a substitution cipher in which the cipher alphabet is fixed through the encryption process. All of the substitution ciphers we have seen prior to this handout are monoalphabetic; these ciphers are highly susceptible to frequency analysis.

Polyalphabetic Cipher. A polyalphabetic cipher is a substitution cipher in which the cipher alphabet changes during the encryption process.

One Time Pads (The Unbreakable)



© 2018 Al-Nafi. All Rights Reserved.

Reference https://en.wikipedia.org/wiki/One-time_pad

In cryptography, the **one-time pad** (**OTP**) is an encryption technique that cannot be cracked, but requires the use of a one-time pre-shared key the same size as, or longer than, the message being sent. In this technique, a plaintext is paired with a random secret key (also referred to as *a one-time pad*). Then, each bit or character of the plaintext is encrypted by combining it with the corresponding bit or character from the pad using modular addition. If the key is (1) truly random, (2) at least as long as the plaintext, (3) never reused in whole or in part, and (4) kept completely secret, then the resulting ciphertext will be impossible to decrypt or break.[1][2] It has also been proven that any cipher with the property of perfect secrecy must use keys with effectively the same requirements as OTP keys.[3] Digital versions of one-time pad ciphers have been used by nations for critical diplomatic and military communication, but the problems of secure key distribution have made them impractical for most applications.

First described by Frank Miller in 1882,[4][5] the one-time pad was re-invented in 1917. On July 22, 1919, U.S. Patent 1,310,719 was issued to Gilbert Vernam for

the XOR operation used for the encryption of a one-time pad.[6] Derived from his *Vernam cipher*, the system was a cipher that combined a message with a key read from a punched tape. In its original form, Vernam's system was vulnerable because the key tape was a loop, which was reused whenever the loop made a full cycle. One-time use came later, when Joseph Mauborgne recognized that if the key tape were totally random, then cryptanalysis would be impossible.[7]

The "pad" part of the name comes from early implementations where the key material was distributed as a pad of paper, allowing the current top sheet to be torn off and destroyed after use. For concealment the pad was sometimes so small that a powerful magnifying glass was required to use it. The KGB used pads of such size that they could fit in the palm of a hand,[8] or in a walnut shell.[9] To increase security, one-time pads were sometimes printed onto sheets of highly flammable nitrocellulose, so that they could easily be burned after use.
There is some ambiguity to the term "Vernam cipher" because some sources use "Vernam cipher" and "one-time pad" synonymously, while others refer to any additive stream cipher as a "Vernam cipher", including those based on a cryptographically secure pseudorandom number generator (CSPRNG).[10]